signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
if real_part > 0 then
phase(i) = atan(imag_part / real_part); // Phase in [-pi/2, pi/2]
elseif real_part < 0 then
phase(i) = atan(imag_part / real_part) + %pi; // Adjust phase for left half-pla
else
if imag_part > 0 then
phase(i) = %pi/2; // Positive imaginary axis
elseif imag_part < 0 then
phase(i) = -%pi/2; // Negative imaginary axis
else
phase(i) = 0; // Origin case
end
end
end
// Plot the magnitude and phase of the DFT
clf; // Clear the current figure
// Plot magnitude
subplot(2,1,1);
bar(k, magnitude, 'b');
xlabel('k');
ylabel('|X(k)|');
title('Magnitude of the DFT');
// Plot phase
subplot(2,1,2);
bar(k, phase, 'r');
xlabel('k');

ylabel('Angle(X(k))');
title('Phase of the DFT');

```

